

Learning Activity Sheet for General Science

Quarter 3

Lesson

8

GOVERNMENT PROPERTY
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Learning Activity Sheet for General Science Grade 11
Quarter 3: Lesson 8

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LEARNING ACTIVITY SHEET

Learning Area:	General Science	Quarter:	3
Lesson No.:	8	Date:	
Lesson Title/ Topic:	Climate Change		
Name:		Grade & Section:	

I. Activity No. 1: What happens if we change?

II. Objective(s): At the end of the activity, you should be able to:

- ✓ Identify at least three key factors (En-ROADS sliders) that influence global temperature changes.
- ✓ Evaluate which factor (slider) has the greatest impact on reducing temperature and justify their reasoning based on simulator results.
- ✓ Explain why global temperature may continue to rise despite the presence of renewable energy sources.
- ✓ Analyze the continued high usage of coal, oil, and natural gas, considering economic, social, and policy factors.
- ✓ Interpret the implications of a projected global temperature increase of +3.3°C and predict possible environmental, social, and economic impacts.

III. Materials Needed:

- ✓ Computers or tablets with internet access
- ✓ Paper or digital devices for note-taking and brainstorming
- ✓ Screenshot or live access to the En-ROADS simulation. Available at <https://en-roads.climateinteractive.org/scenario.html?v=25.10.0>

III. Instructions:

Show the class the En-ROADS screen with all sliders at status quo or provide them a copy of the screenshot of En-ROADS interface like the one provided below. Ask learners to observe and take down notes.

Guide Questions:

1. What is the projected temperature increase by 2100?
2. Which energy sources dominate (coal, oil, gas, renewables, etc.)?
3. What do the greenhouse gas emissions graph and numbers tell us?

Then, show the class how the simulator works. Explain each part and how slider works. Adjust the sliders in the simulator according to learners' suggestions and observe the impact on temperature and emissions.

Afterwards, ask them to form small groups. Each group will choose three sliders they would adjust to reduce the temperature increase.

Processing Questions:

1. What are the three factors (sliders) you used to reduce the temperature?
2. What was the most effective factor (slider) in reducing the temperature? Why do you think so?

Additional questions:

1. Why do you think the temperature continues to rise even though renewable energy is present?
2. What could be the reasons coal, oil, and natural gas usage remain high?
3. What does a +3.3°C increase mean for the planet? What impacts might we expect?

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IV. Activity No. 2: *Exploring Climate Change*

V. **Objective(s):** At the end of the activity, you should be able to:

- ✓ Explain the role of carbon dioxide (CO₂) in influencing atmospheric temperature.
- ✓ Describe the relationship between increasing atmospheric temperature and the amount of water vapor in the air.
- ✓ Analyze the impact of increasing human emissions on global temperature.
- ✓ Interpret how ice growth or melting is affected by increased human emissions.
- ✓ Identify and explain patterns of interaction between temperature, CO₂ levels, water vapor, and ice coverage over time using simulation data.

III. Materials Needed:

- ✓ Computers or tablets with internet access
- ✓ Paper or digital devices for note-taking and brainstorming
- ✓ Climate Change Simulator entitled “High Adventure Science: Exploring Climate Change” by The Concord Consortium. Available at <https://has.concord.org/global-climate.html>

VI. Instructions:

Explore the climate change simulator entitled “High Adventure Science: Exploring Climate Change” by the Concord Consortium and answer the following questions:

Guide Questions:

1. What role does CO₂ play in affecting atmospheric temperature?
2. How does an increase in atmospheric temperature influence the amount of water vapor in the air?
3. What happens to temperature when you increased the human emissions by 150% and 200% respectively?
4. Describe how the ice grows or melts as the human emissions is increased.
5. Using the model, what patterns do you observe in how temperature, CO₂, water vapor, and ice interact over time?